EXTRA PROBLEMS
For THE MEAN-VALUE THEOREM

Problem: Let a function f be defined on $[1,2]$. Show that f does not satisfy the condition of the MVT. Does the conclusion of the theorem hold in this case?

$$f(x) = \begin{cases} 1 & \text{if } x = 1 \\ x^2 & \text{if } 1 < x < 2 \\ 2 & \text{if } x = 2 \end{cases}$$

Sol. Since the function is discontinuous at $x=2$. Because $\lim_{x \to 2} f(x) = 4 \neq f(2)$. So, the first condition of MVT is violated.

Note that the conclusion is also not true.

Since $\dfrac{f(2)-f(1)}{2-1} \neq f'(c)$ for any $c \in (1,2)$.

Because, $f'(c) = 2c$, $f'(x) = 2x \quad \forall x \in (1,2)$

and $\dfrac{f(2)-f(1)}{2-1} = \dfrac{2-1}{2-1} = 1 \neq f'(c)$

$\square$

Problem: Let $f(x) = |x| \ \forall x \in [-1,2]$. Does f satisfy all the conditions of the MVT? Does the conclusion of the theorem hold? Justify your answer.

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* The condition of the MVT are only sufficient. They are not NECESSARY. This is evident from the following function.

$$f(x) = \begin{cases} 0 & \text{if } 0 \le x < \frac{1}{4} \\ x & \text{if } \frac{1}{4} \le x < \frac{1}{2} \\ \frac{x}{2}+1 & \text{if } \frac{1}{2} \le x \le 2 \end{cases}$$

You can see that $\frac{3}{8}$ lies in $(0,2)$ and $f'(\frac{3}{8}) = 1 = \frac{f(2)-f(0)}{2-0}$

However, f is neither continuous on $[0,2]$, nor differentiable on $(0,2)$.

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